# Transformer-BEATS: A Transformer model for time series prediction of dairy milk production.

Vahid Naghashi* Mounir Boukadoum and Abdoulaye Banire Diallo
Department of Computer Science, Université du Québec à Montréal , Montreal, Canada
naghashi.vahid@courrier.uqam.ca

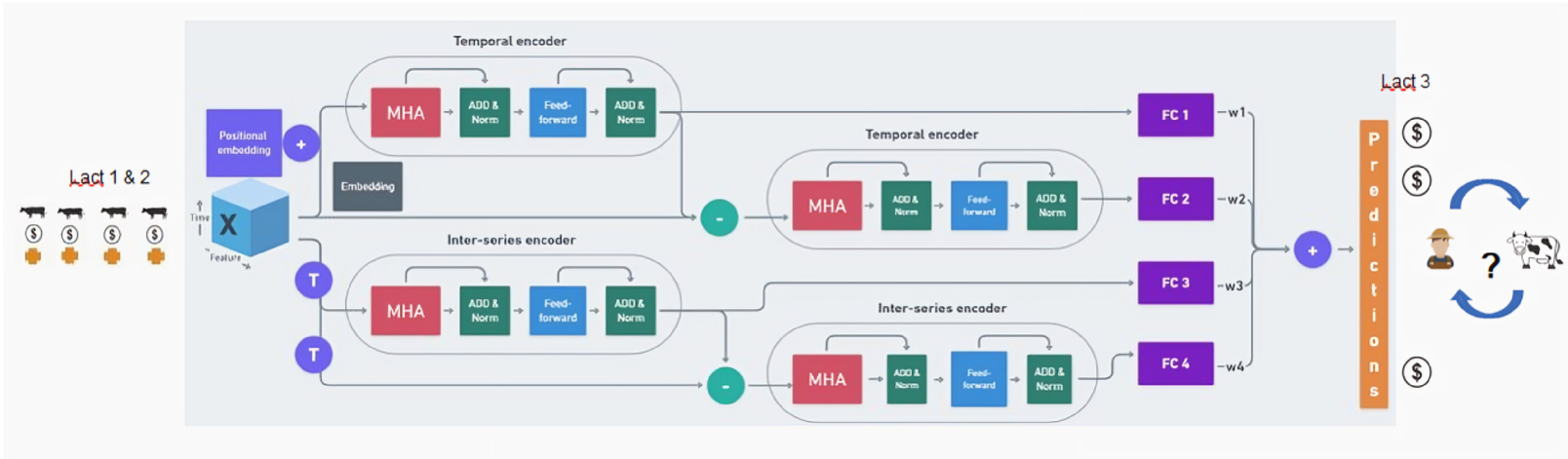
## 1. Introduction

Predicting the milk production of dairy cows is important for precision livestock management. It can be achieved by observing the historical volume of each cow and features encoding its health status and the farm environment. This problem can be modeled as a multivariate time-series forecasting task, which demands capturing temporal dependencies and inter-series relations effectively. The current time-series models for dairy forecasting often fall short to capture these intricate temporal dependencies and inter-series (spatial) relationships. Here, we present a Transformer-based architecture that effectively captures these spatial and temporal dependencies through self-attention components, with a new adaptive way to integrate the forecasts from the temporal and spatial encoders. The proposed model is enhanced by connecting two Transformer encoder blocks in a back-cast (residual) manner, ensuring that the most relevant information is provided to the following block. Our approach yields better results, particularly in terms of average prediction error per cow, than the state-of-the-art methods when predicting milk production in upcoming lactation of 47,000 cows.

## 2. Problem Definition

Prediction of milk production can be stated as a time series forecasting task: given a herd of cows with $N$ samples, each cow $i$ represented by a set of records $\{\mathbf{x}_j^{(i)} = (x_{j,1}^{(i)}, ..., x_{j,T}^{(i)})\}_{j=1..v} \in \Re^{v \times T}$, where $\mathbf{x}_j$ is a sequence of dairy variables, $v$ representing the number of dairy variables considered and $T$ denoting the record length (total length of the first and second lactation months, fixed for all of the animals), we aim to predict M steps (months) of the upcoming milk production within the third lactation: $\hat{r}^{(i)} \in \Re^M$.

## 3. Transformer-BEATS model




## 4. Inter-series Attention Block

In time series forecasting, the dependencies between various channels is informative in the prediction process. In the projection formulas below, $X$ is the transposed input multivariate series without embedding of shape $(Batch, Channel, SequenceLength)$. $W_{int}^Q$, $W_{int}^K$ and $W_{int}^V$ are of dimensions $d_T * d_T$ with $d_T$ indicating the time dimension size (number of the input time steps).

$$Q_{int} = X \cdot W_{int}^Q, K_{int} = X \cdot W_{int}^K, V_{int} = X \cdot W_{int}^V \quad (1)$$

The inter-series attention scores are calculated in the same way as in the standard self-attention:

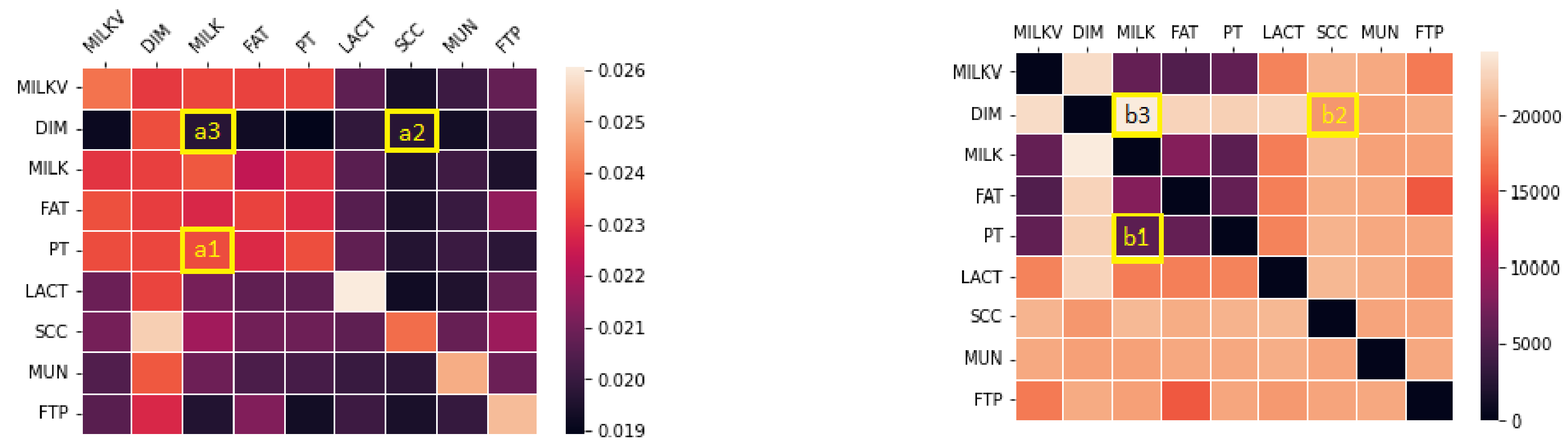
$$Score_{int} = softmax\left(\frac{Q_{int} \cdot K_{int}^T}{\sqrt{d_T}}\right) \quad (2)$$

$$Attention(Q_{int}, K_{int}, V_{int}) = Score_{int} \cdot V_{int} \quad (3)$$

## 5. Experimental Results

Forecasting results related to longer horizon prediction in the dairy production dataset (Input sequence length = 11 months and target sequence length = 22 months).

| **Method** | Cum-RMSE | Cum-MAE |
|---|---|---|
| Transformer-BEATS | **54.417** | **42.716** |
| DLinear | 56.978 | 44.783 |
| MuMu | 55.024 | 43.198 |
| iTransformer | 57.118 | 44.814 |
| Crossformer | 54.623 | 42.780 |
| PatchTST | 57.613 | 45.666 |




**Left**: Mean inter-series (spatial) attention map learned from the data samples (Numerical features). **Right**: Mean channel-wise dynamic time warping distance matrix of the test samples. Blocks b1, b2, and b3 indicate low, high, and relatively high DTW distance between the dairy feature pairs, while blocks a1, a2, and a3 represent high, low, and low attention scores, respectively. A low DTW distance reflects a high attention score, and vice versa.

## 6. Conclusions

We propose Transformer-BEATS, a Transformer based framework for prediction of the multivariate time series of the dairy milk income in upcoming lactation months.

1. Our model captures both temporal and cross-channel dependencies in the dairy time-series data by combining different Transformer encoders.
2. The encoder blocks are inter-connected in a residual (back-cast) manner to help useful signal propagation to the next block.
3. Transformer-BEATS achieves the best results in terms of the error metrics.
4. We applied Transformer-BEATS to longer horizons and attained promising results.
5. The proposed model will be extended to forecast different lactation cycles.

## 7. References

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